

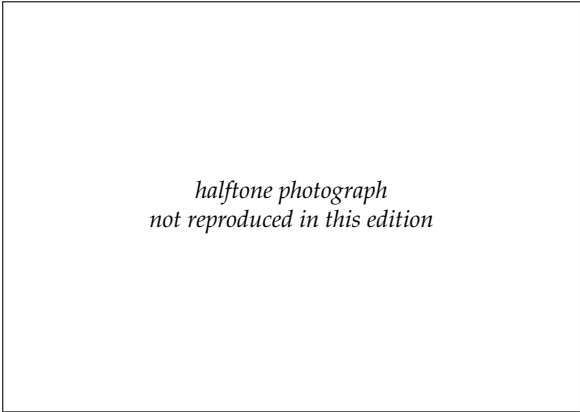
# GEOMETRY

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# GEOMETRY

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ADDISON-WESLEY PUBLISHING COMPANY, INC.

READING, MASSACHUSETTS, U.S.A.

LONDON, ENGLAND

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*Library of Congress Catalog Card No. 60-8336*

Second printing, February, 1961



## PREFACE

It is a common belief that plane geometry was completed by Euclid 2000 years ago and that nothing has been added to it or taken from it since. This is simply not true. That there are logical gaps in Euclid's presentation has been known for a long time. Means to remedy these deficiencies have been known for about sixty years, but strangely enough a mathematically adequate and yet elementary treatment of plane geometry in the spirit of Euclid has not appeared in print. This text represents an earnest effort to do just this. The current interest in improvement of the secondary school curriculum makes this an appropriate time for such an attempt.

This book, in earlier unpublished versions, has been taught to *normal* classes in *average* high schools for five years. Many of the classes were taught by one of us in the Burris School, the laboratory school of Ball State Teachers College, but the material has been taught in other schools by teachers who had had no previous special training in geometry. Our experience is that both student and teacher find this treatment of geometry stimulating. The good students become excited about mathematics and the students of average ability learn the basic geometric facts.

Besides its mathematical completeness, an attribute that makes for clarity and ease of understanding, there are several features which we think important.

There is a full chapter on logic early in the text. It pays the teacher to return to this logic again and again in order to clarify

the principles underlying deductive proof. For the nonscience student, this chapter on logic may be the item of greatest permanent value.

The postulates are presented in groups, as needed, with a resulting gain in clarity and simplicity. Although it is true that at the start most students are hardly “provers,” by the end of the year a considerable part of the class will exhibit real mathematical perception. In the early stages the small number of postulates makes proofs easier to construct.

Definitions are given explicitly; that is, they are formally labeled as such. It is one of our prime objectives to get the student to see that precise definitions are necessary for proofs.

There is plenty of material for a full year’s work for all kinds of classes. More difficult sections, theorems, or exercises are marked with an asterisk, and some of the more difficult theorems may be treated as postulates. The chapters on analytic geometry, locus, solid geometry, and philosophy of mathematics provide supplementary material beyond the standard course. The chapter on solid geometry alone could occupy a good portion of a semester if desired.

There is a review of important concepts at the end of each chapter. In addition, there are algebra review sections, designed not only to refresh skills but also to bring out the logical structure of algebra.

The Appendix lists all the postulates, plus the principal definitions and theorems. Besides being a convenient reference, the Appendix is worth studying during the year, for it gives, in brief, the step-by-step construction of the full geometry.

We believe that the text raises students’ reading ability. There is little verbiage. Each section has a definite purpose, and the text is directed to this purpose. Good classroom discussion, with careful attention to the use of words, will add much to each student’s ability to express himself clearly.

A separate Teachers’ Manual gives greater detail on selection



of topics, emphasis, and time schedule. There is also an Answer Book for the exercises.

It is a great pleasure to thank all those who have helped us with encouragement or constructive criticism. Special thanks are due to Dr. Curtis Howd, Principal of the Burris Laboratory School, who encouraged us and made possible the experimental sections; to Professor Jack Forbes, who observed many of the classes in various schools and who insisted that the mathematics be correct; to teachers at the Burris School, and many others, who by teaching the material have improved the exposition; to The National Science Foundation, which, through a grant for in-service institutes at Ball State, enabled us to reach a wider audience and see the text taught by many teachers in a variety of school situations; and finally, to our wives, who have waited impatiently for the publication day.

*January 1960*

C. B., R. E., M. S.



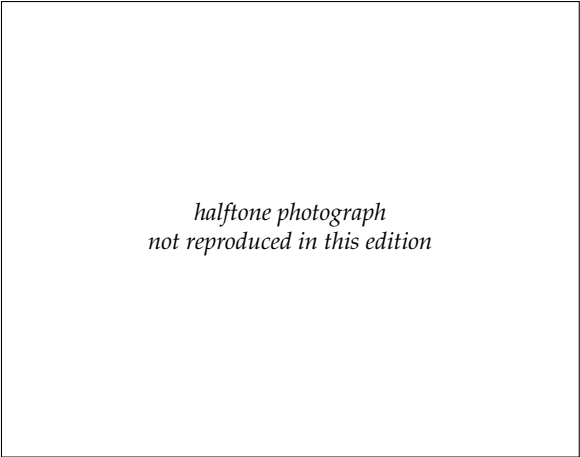
## FOREWORD TO THE STUDENT

Mathematics is much more than a collection of interesting facts and useful tools for solving special problems. Mathematics is primarily concerned with the *development* of important ideas. Frequently mathematicians develop an interesting mathematical theory that is apparently useless so far as the solution of practical problems is concerned. Later someone finds that this mathematics fits very nicely into the pattern of some complicated practical problem. The mathematical theory then becomes a useful tool.

As a branch of mathematics, geometry is first of all concerned with the development of geometrical theory. As you study geometry you will see again and again how the geometrical concepts fit into patterns in the real world and become helpful in the solution of practical problems.

Mathematics requires thought, and thinking deserves your best effort. Do not hesitate to ask questions.

Good luck,  
*The Authors*



*halftone photograph  
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DAVID HILBERT  
(1862–1943)

Professor of Mathematics,  
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Perhaps the world's foremost mathematician during the  
first half of the twentieth century.

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